

# Barriers to International Student Mobility

Survey Analysis – Four-Group Approach



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The Survey

Who Are the Four Groups?

Measuring Overall Constraint

What Predicts Group Membership?

Conclusions



## Sample

- **1230** total respondents
- **808** usable ( $\geq 70\%$  of key variables)
- Median completion rate: **68%**

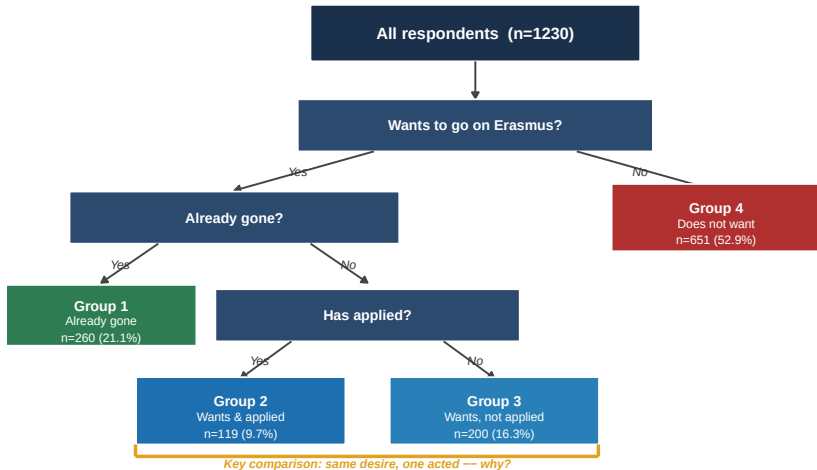
## Four Groups

1. Already gone on Erasmus
2. Wants to go & has applied
3. Wants to go, has not applied
4. Does not want to go

## Variables

- 16 barriers (Likert 1–5)
- 11 reasons for going
- 27 psychological profile items
- Socio-demographic variables
- **New:** Parental Erasmus (Q13), Certified English (Q24)

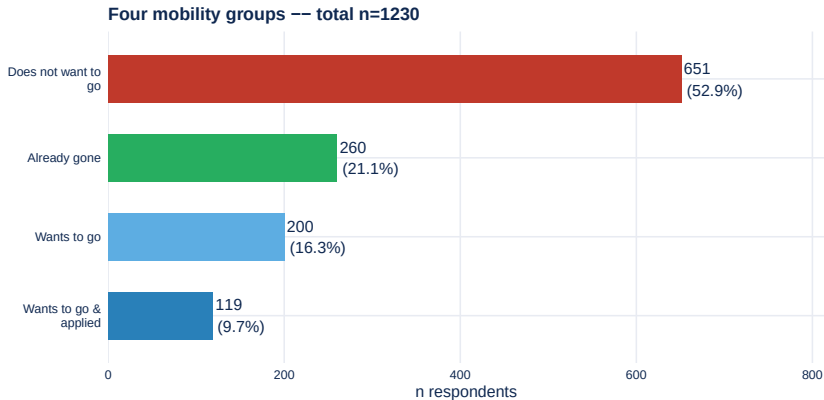
# How Are the Four Groups Defined?



*Decision tree: each student is assigned to exactly one group based on three binary questions. Groups 2 and 3 share the same desire for mobility – the key analytical question is what differentiates them.*

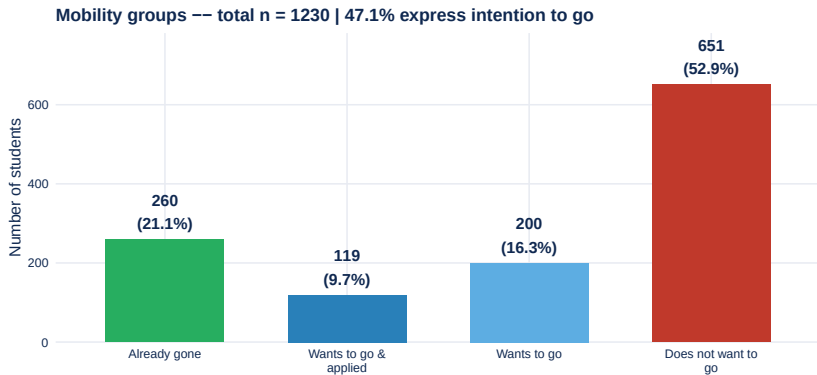


# The Four Mobility Groups



*Reading: each bar = number of students. Focus on Groups 2–3 (both want to go): the gap reveals how many are blocked at the application stage.*

# Departure Rates and Conversion Funnel



*47.1% of respondents express some intention to go abroad. Among those who want to go, the conversion to application is the key lever studied.*



## Two Central Questions

- **Who** are the four groups?  
(socio-demo, academic, financial, psychological)
- **What** differentiates students who applied from those who want to go but haven't?

## Three Mechanisms

1. **Structural:** finances, grades, parental capital
2. **Psychological:** international profile, openness
3. **Informational:** lack of knowledge about procedures

## Key Comparison

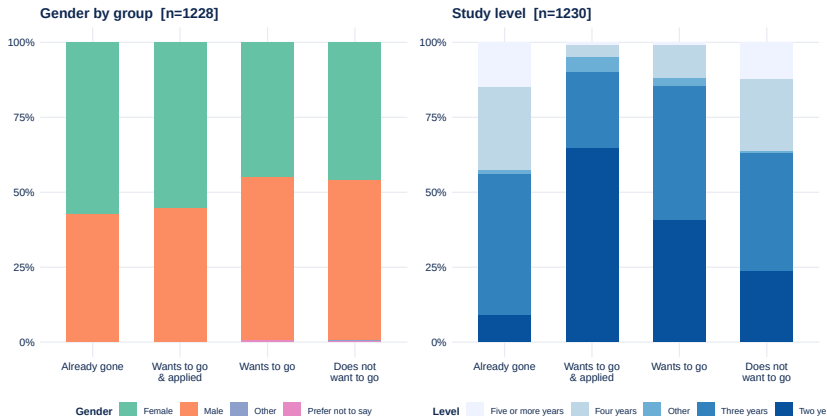
Groups 2 vs 3: same desire, but only one has taken action.

**What makes the difference?**

## Methods

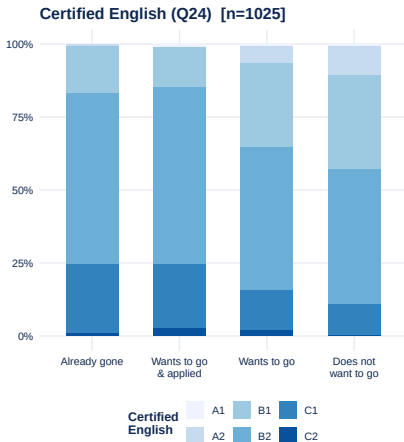
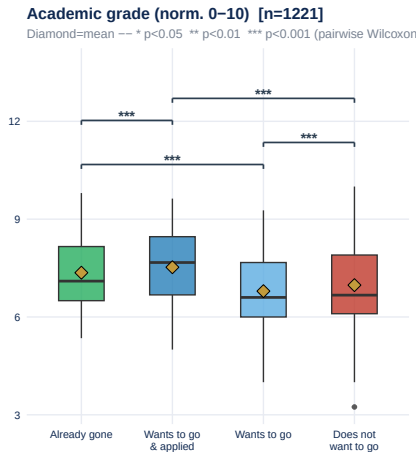
- Descriptive by group
- MCA + composite barrier score
- OLS on score
- Logistic regressions

# Socio-Demographic Profile by Group



*Reading: stacked bars show group composition. Look for visible differences in gender or study-level distribution – these signal structural selection into mobility.*

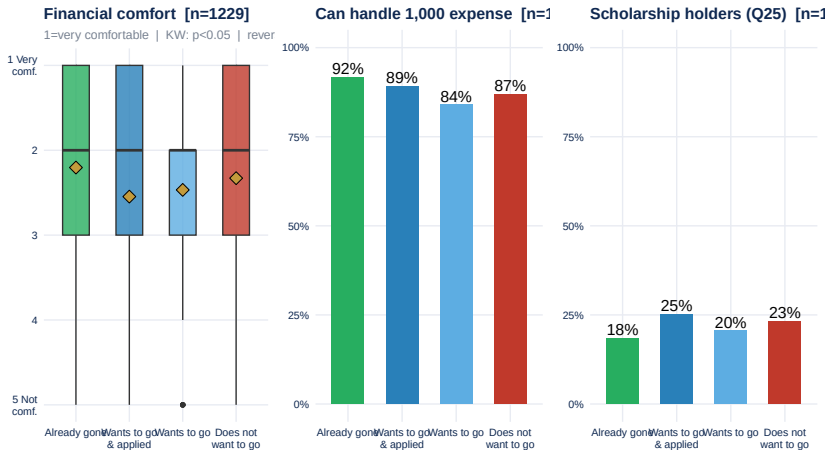




*Reading: box = middle 50% of values, diamond = mean. A higher mean grade in Group 1 (already gone) would suggest academic selection into Erasmus participation.*



# Financial Profile by Group

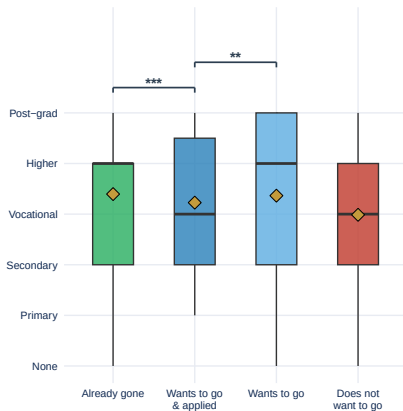


Left: financial comfort scale reversed (1=very comfortable). Kruskal-Wallis global test reported in subtitle. Significant pairs (Bonferroni): Already gone vs Wants to go \*. Middle: share able to absorb a 1,000 unexpected expense. Right panel: share of scholarship holders (Q25) by group – higher scholarship rates in constrained groups would confirm structural financial barriers

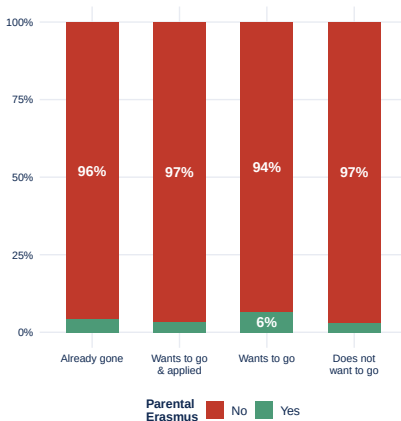
# Parental Capital by Group

## Parents' education level

\*  $p < 0.05$  \*\*  $p < 0.01$  \*\*\*  $p < 0.001$  (pairwise Wilcoxon, Bonferroni)



## Parental Erasmus participation (Q13)



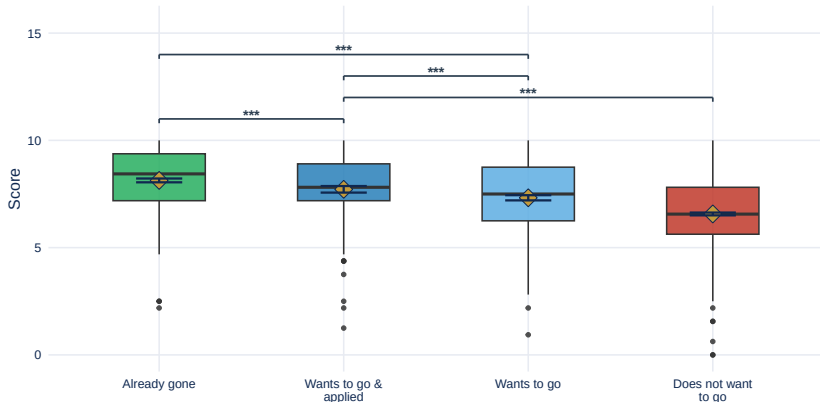
*Reading: left = parents' education level (higher = more cultural capital). Right = parental Erasmus experience provides direct knowledge of the programme and process.*



# International Profile by Group

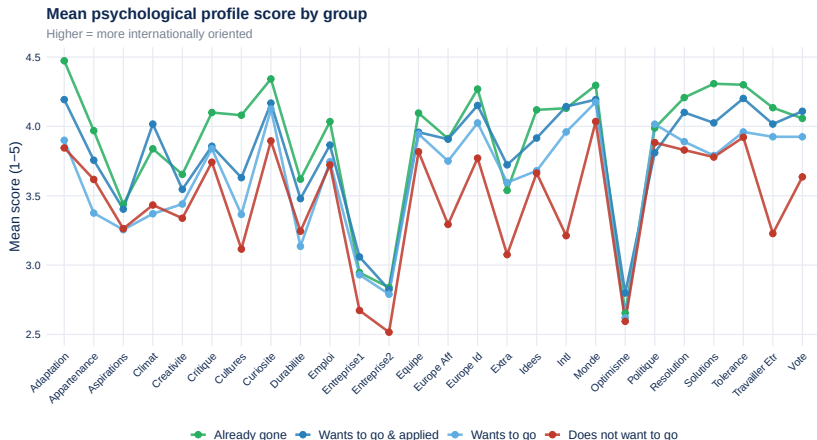
## International profile score (0–10) by group

Diamond = mean  $\pm$  SE | KW:  $p < 0.001$  | \*  $p < 0.05$  \*\*  $p < 0.01$  \*\*\*  $p < 0.001$  (Wilcoxon, Bonferroni)



*Kruskal-Wallis across 4 groups:  $p < 0.001$ . Pairwise brackets show Bonferroni-corrected Wilcoxon tests (significant pairs only). Higher score = more open to international mobility.*

# Psychological Profile: Group Differences



*Reading: each line traces mean scores across 27 psychological dimensions. Groups 1-2 scoring higher on openness/international dimensions than Group 4 would confirm the psychological mechanism.*

## Are Group Differences Statistically Significant?

Table 1: Group differences: statistical tests

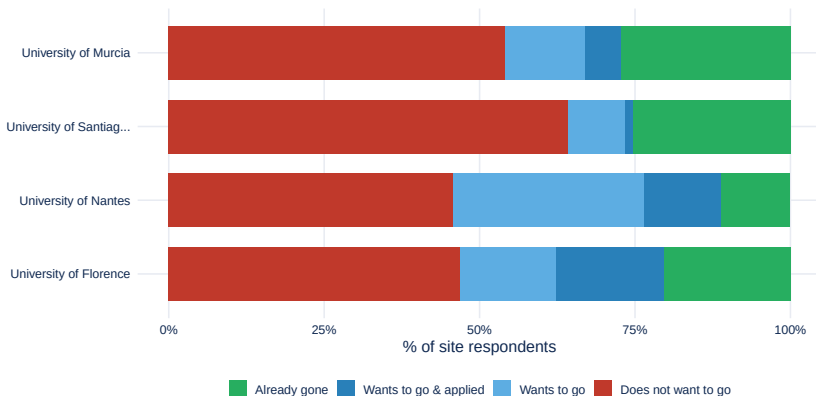
| Variable          | Test           | Statistic | p-value | Sig. |
|-------------------|----------------|-----------|---------|------|
| genre             | Chi-squared    | 17.79     | 0.041   | Yes  |
| niveau            | Chi-squared    | 207.18    | 0.000   | Yes  |
| zone_geo          | Chi-squared    | 43.26     | 0.017   | Yes  |
| parental_erasmus  | Chi-squared    | 4.96      | 0.173   | No   |
| moyenne_acad_norm | Kruskal-Wallis | 54.00     | 0.000   | Yes  |
| aisance_fin       | Kruskal-Wallis | 10.12     | 0.018   | Yes  |
| score_intl        | Kruskal-Wallis | 193.90    | 0.000   | Yes  |
| educ_num          | Kruskal-Wallis | 25.93     | 0.000   | Yes  |

*Reading:  $p < 0.05$  = group difference is statistically reliable. n.s. = not significant with this sample size – not the same as no effect.*

# University Effect: Group Composition by Site

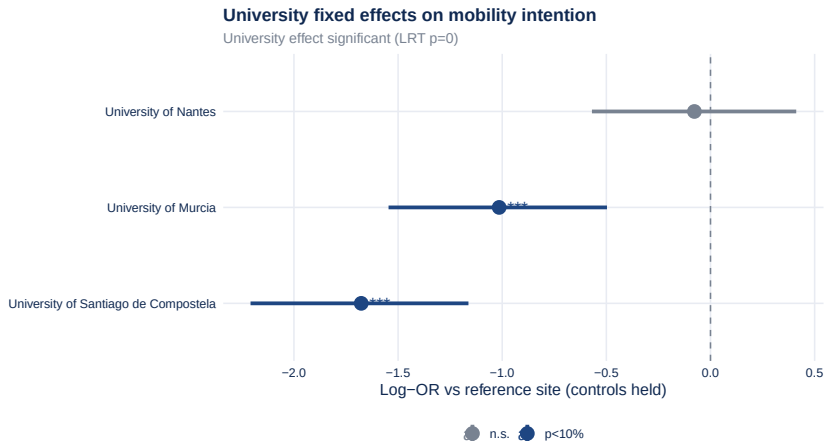
## Group composition by university (n=4 sites)

Rows sum to 100% -- compare 'Wants to go' share across sites



*Stacked bars show the share of each mobility group per site. Sites with a larger 'Already gone' share may have a stronger Erasmus culture or more systematic information. Compare the 'Does not want to go' share to identify sites with higher structural constraint.*

# University Effect: Fixed Effects Logit



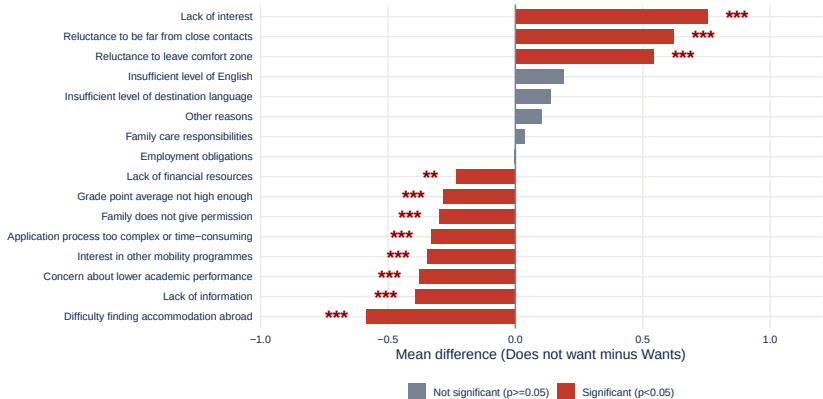
*University fixed effects after controlling for financial comfort, academic grade, parental education and international profile score. A significant LRT ( $p < 0.05$ ) means the site matters independently of who studies there. Coefficients are log-OR relative to the reference university (lowest alphabetical). Stars: \*  $p < 10\%$ , \*\*  $p < 5\%$ , \*\*\*  $p < 1\%$ .*



# Which Barriers Differentiate Wants vs Does Not Want to Go?

## Barriers: Does not want vs Wants to go [n=970]

Wants to go n=319 x Does not want n=651 | \* p<0.10 \*\* p<0.05 \*\*\* p<0.01 (Mann-Whitney)

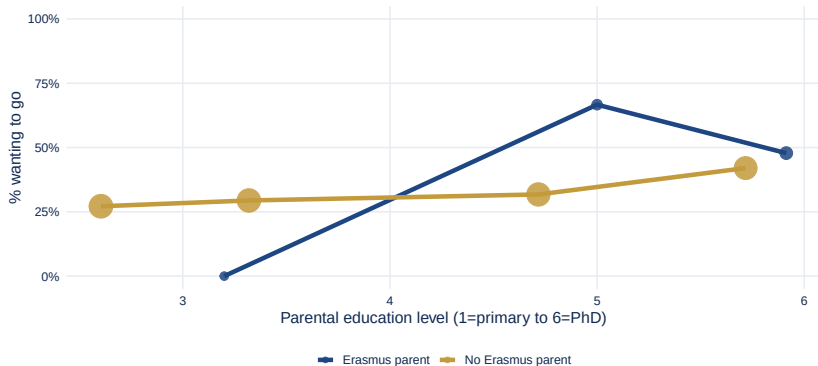


*Positive bars = barriers rated higher by students who do NOT want to go. Red bars (starred) are statistically significant (Mann-Whitney  $p < 0.05$ ): 11 out of 16 barriers. These are the barriers that genuinely differentiate the two groups.*

# Interaction: Parental Erasmus x Parental Education

## Parental education x Parental Erasmus → Mobility intention

Interaction n.s. ( $p=0.301$ ) -- Logit:  $y \sim \text{educ}(z) * \text{parental\_erasmus}$  --  $n=970$



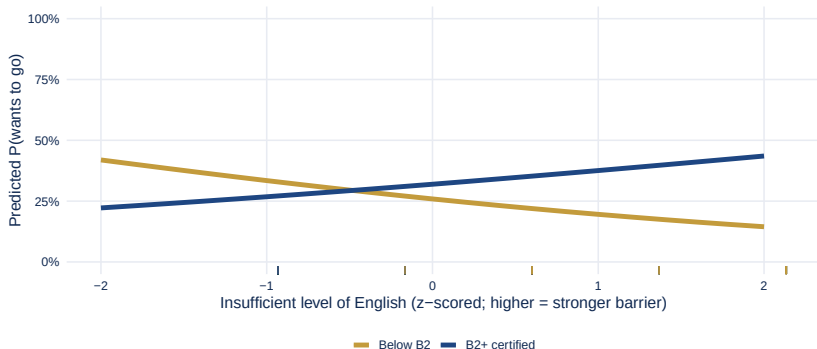
*Lines show % of students wanting to go by parental education quartile, split by parental Erasmus experience. A significant interaction ( $p < 0.05$ ) would mean the education gradient is steeper (or flatter) when a parent has done Erasmus – distinguishing general cultural capital from programme-specific knowledge. Point size = cell n.*

# Interaction: Certified English x Language Barrier

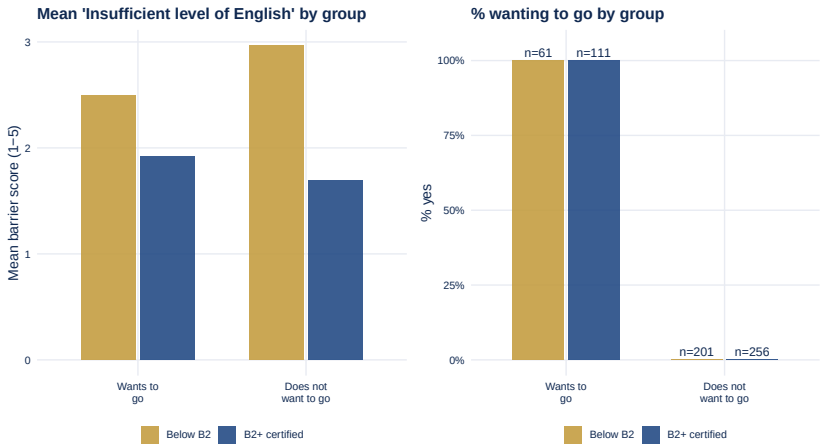
## English level x Language barrier → Mobility intention

Interaction significant ( $p=0.002$ ) | B2+  $n=367$ , Below B2  $n=262$

Barrier: Insufficient level of English



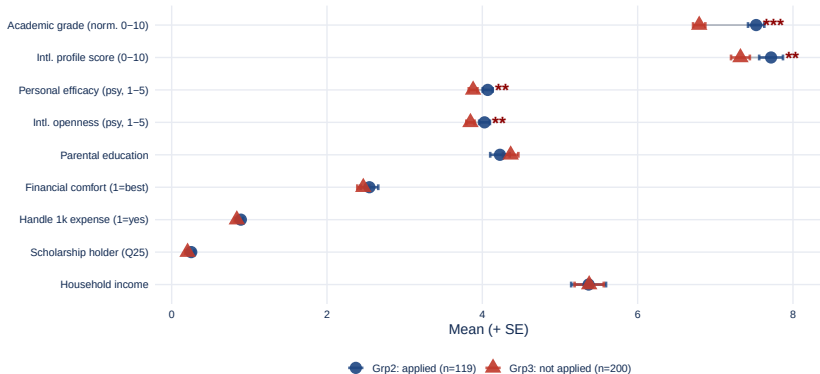
*Predicted probability of wanting to go as a function of perceived language barrier, split by English certification status (B2+ vs below B2, Q24). A significant interaction confirms that certification attenuates the language barrier: certified students show a weaker negative slope between perceived language difficulty and mobility intention.*



Left: mean perceived language barrier score (1=no obstacle, 5=major obstacle) by mobility group and certification level. Right: % wanting to go by group and certification level. If certified students perceive a lower barrier AND show higher mobility intention, the certification is doing double duty – both reducing perceived difficulty and boosting confidence.

## Profile comparison: Group 2 vs Group 3 [n=319]

Grp2 n=119 vs Grp3 n=200 | \*  $p < 0.10$  \*\*  $p < 0.05$  \*\*\*  $p < 0.01$  (Mann-Whitney)

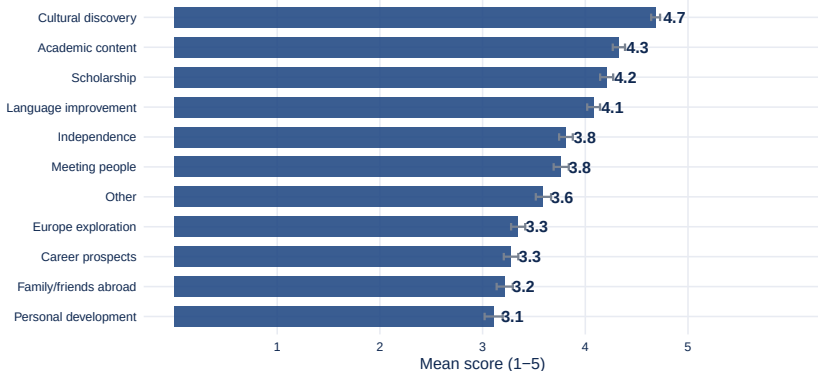


*frein\_ items excluded (NA for Grp2 by design). Blue circle = Grp2 (applied), red triangle = Grp3 (not applied). 4 variable(s) significantly different at  $p < 0.05$  (Mann-Whitney).*

## Reasons for Going – Group 1 (Already Gone)

### Reasons for going -- Group 1: Already gone [n=260]

1 = not at all important | 5 = very important | bars = mean  $\pm$  SE

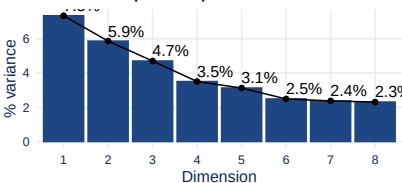


*The 'reasons for going' question (raison\_) is answered only by Group 1 (already gone). Other groups (2, 3, 4) did not answer this question. To compare motivations between Groups 2 and 3, refer to the barrier gap plots.*

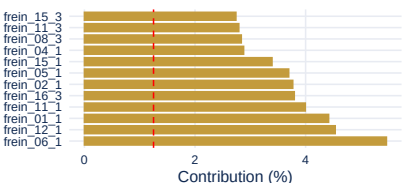


# MCA on Barriers

Variance explained per dimension

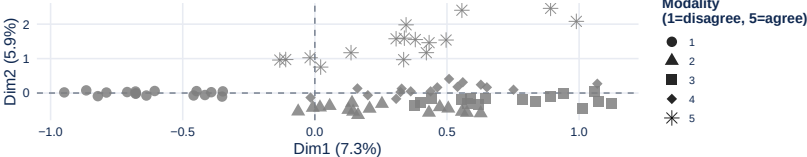


Contribution of barriers to Dim1



MCA -- modality map (Dim1 x Dim2)

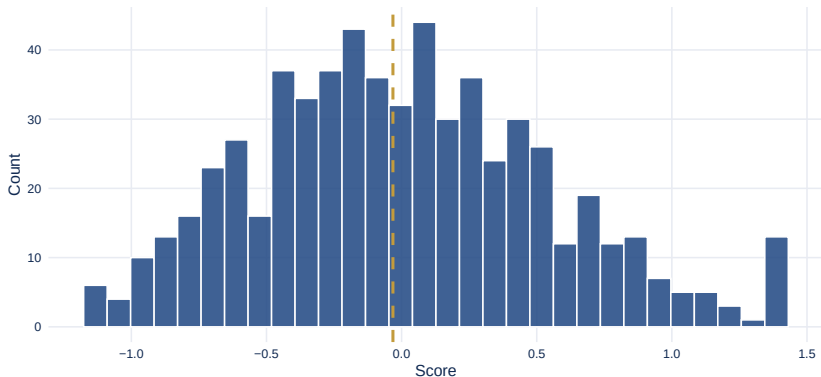
Dim1: 7.3% -- Dim2: 5.9% of variance



# Composite Barrier Score

## Composite Barrier Score -- Option B (MCA Dim1)

Dashed = median -- n = 613 | Composite barrier score (MCA Dim1, recentred)



## Scope limitation

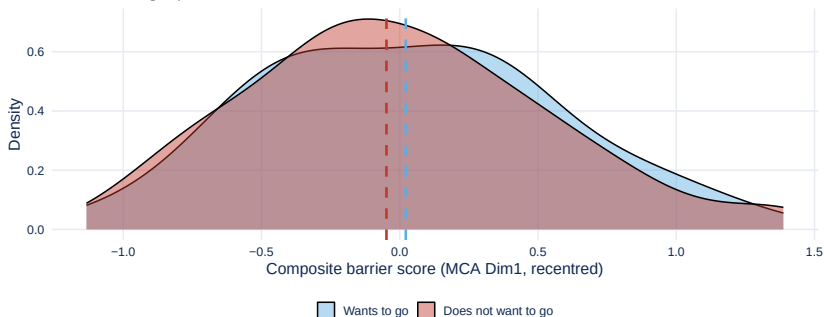
The barrier items (`frein_`) were not administered to **Group 2** (applied,  $n = 119$ ) due to questionnaire routing: **100%** of their scores are NA. All score-based analyses (distribution, OLS, KW) therefore cover **Groups 1, 3 and 4 only**. Group 2 appears in group-level plots as an empty or near-empty curve.



# Composite Barrier Score – Distribution

## Composite barrier score -- Option B (MCA Dim1)

Dashed lines = group medians



*Reading: dashed lines = group medians. Curves shifted right = more barriers felt. Group 2 (applied) is absent from this plot (100% NA on barrier items). Groups 1, 3 and 4 are the analytic base for all score-based comparisons.*

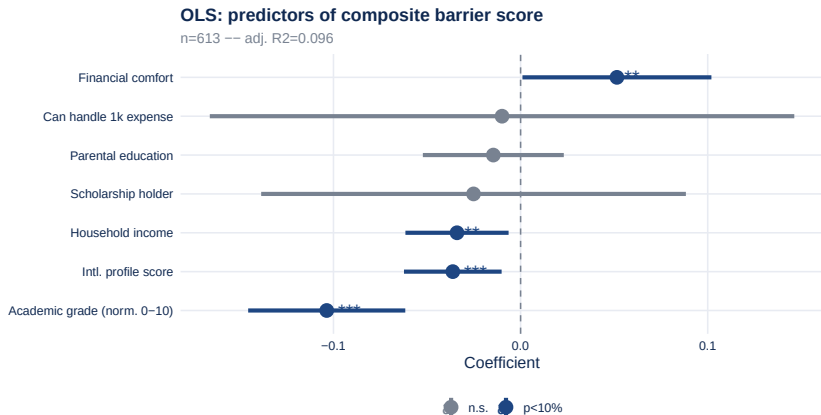
# Score by Group: Does It Separate the Four Groups?



*Kruskal-Wallis across 4 groups:  $p=0.496$ . Higher score = more barriers felt.*

**Caution:** *Group 2 contributes only 0% non-missing scores — its box is near-empty and should not be interpreted.*

# What Socio-Demo Variables Predict the Score?

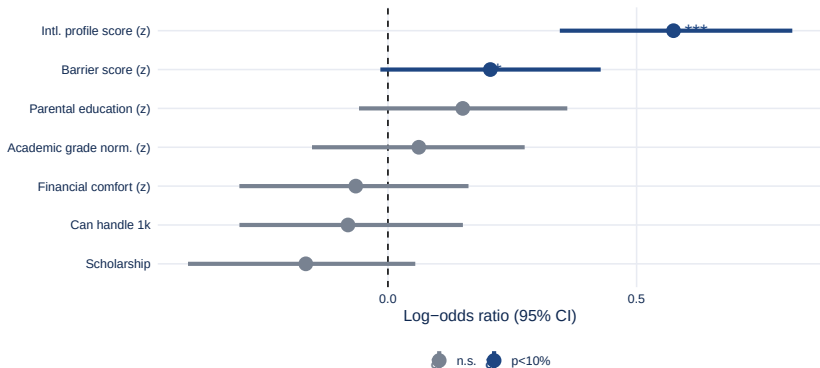


*Reading: bars right = variable increases barrier score. Stars = significance (\*  $p<10\%$ , \*\*  $p<5\%$ ). Financial comfort is coded 1 = very comfortable to 5 = not comfortable: its positive coefficient means students who are less financially comfortable perceive more barriers. Group 2 is excluded de facto (100% NA on the dependent variable).*

# Logistic Regression: Movers vs Non-Movers

## Logit: who wants to go? (Yes vs No)

Log-OR, standardized predictors (z) | n=607



*Reading: % change in odds of wanting to go. International profile is typically the dominant predictor. Significant barriers with negative effect actively deter mobility intention.*

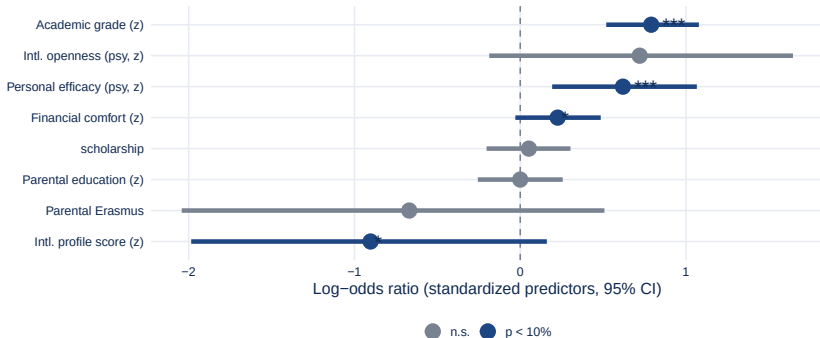
# Applied vs Did Not Apply (Groups 2 vs 3)

## What drives the decision to apply? (Logistic regression)

Groups 2 vs 3 | complete cases n=315 (applied: 118, not applied: 197)

Predictors standardized (z). Barrier score excluded (NA for Grp2).

Raw: Grp2=119, Grp3=200



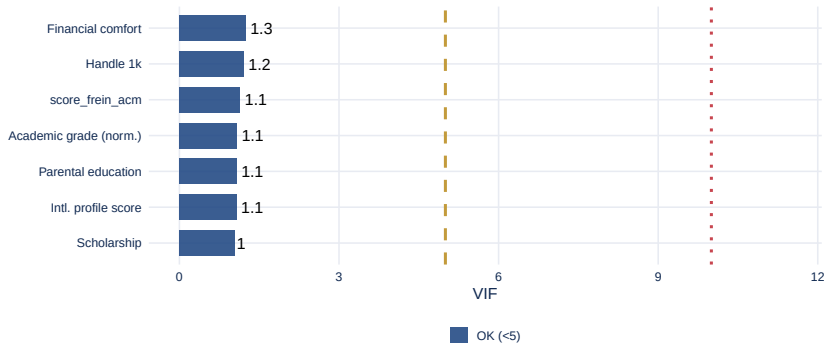
*Reading: % change in odds of applying among students who want to go.*

*Barrier score: Method A (simple mean, robust to partial NA). Same desire, different action – what tips the balance towards applying?*

# Multicollinearity Check (VIF)

## Variance Inflation Factors -- Movers logit

Dashed = VIF 5 (moderate) -- Dotted = VIF 10 (severe) -- Values < 5: no issue

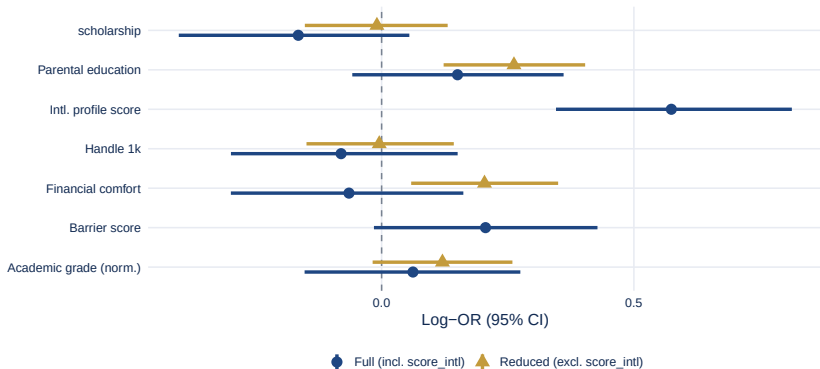


*VIF (Variance Inflation Factor) measures how much a predictor's variance is inflated by collinearity with others.  $VIF > 5$  = moderate concern;  $VIF > 10$  = severe. Here: 0 predictor(s) with  $VIF \geq 5$ , 0 with  $VIF \geq 10$ . Financial variables (*aisance\_fin*, *revenu\_num*, *depense\_imp*) are the most likely collinear cluster; removing one or replacing all three with the structural vulnerability composite is recommended if  $VIF > 5$ .*

# Robustness Check: Reduced Model

## Robustness: full vs reduced movers logit

LRT not available -- Stable coefficients = no masking by collinearity

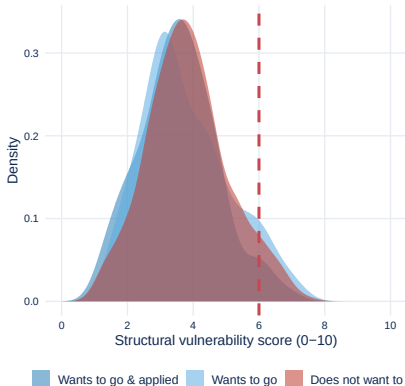


Blue = full model; gold = reduced model without `score_intl` (international profile score). If structural coefficients (financial comfort, parental education, grade) change substantially between models, `score_intl` was masking them. Structural coefficients (financial, parental, academic) are stable across specifications. If the LRT is n.s., `score_intl` may be dropped without loss of explanatory power for structural effects

# Vulnerability Score: Validation

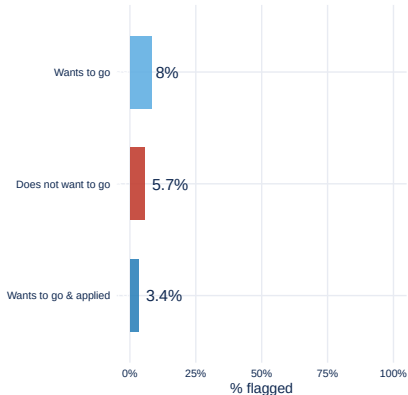
## Vulnerability score distribution by group

Dashed = threshold  $V > 6$  (resigned non-mover flag)



## % with $V > 6$ by group

If score is valid, Grp4 should have highest share



| Variable                    | $V > 6$ | $V \leq 6$ | n ( $V > 6$ ) |
|-----------------------------|---------|------------|---------------|
| Financial comfort (1=best)  | 3.58    | 2.31       | 57            |
| Academic grade (norm. 0-10) | 6.28    | 7.05       | 56            |
| Parental education (1-6)    | 2.77    | 4.18       | 57            |
| Intl. profile score (0-10)  | 7.04    | 6.85       | 57            |
| Scholarship holder (%)      | 40.1%   | 21.2%      | 57            |



## Methodological limits

- **Cross-sectional design:** associations only, no causal inference. Psychological profile may be shaped *by* the experience of applying, not only precede it.
- **Self-selection bias:** voluntary participation likely under-represents the most disengaged students (Grp4), biasing barrier scores downward.
- **Group 2 size:** if  $n < 20$ , logistic regression estimates are unstable — interpret with caution.
- **Barrier items unavailable for Grp2:** all score-based comparisons exclude applied students by design.

## Measurement limits

- **Measurement invariance not tested:** psy items assumed to measure the same constructs equally across groups.
- **Likert interval assumption:** profile scores treat 1–2–3–4–5 as equally spaced; ordinal modelling not implemented.
- **Vulnerability weights (0.4/0.3/0.3):** theoretically motivated, not empirically derived; no sensitivity analysis.
- **Grade normalisation:** assumes students report grades on their university scale — no cross-check available.

## Scope

Results valid for the surveyed sample

Replication across additional cohorts

## 3 Mechanisms Confirmed

- **Structural:** financial comfort, parental capital, grades predict barrier score and group membership
- **Psychological:** international profile score is the strongest predictor of mobility intention
- **Informational:** lack of knowledge about procedures is independently significant

## Key Finding

Grp2 vs Grp3 (applied vs not): gap driven more by **psychological profile** and **information access** than by financial constraints.

## 3 Action Levers

- **Info.** Clear guides + Erasmus testimonials
- **Psych.** Intercultural events from Year 1
- **Struct.** Emergency grants for constrained profiles

## Limitation

$n = 1230$  respondents. Group 2 may be small — interpret with caution. Data collection ongoing.

### Group 3 → Group 2 (lever priority)

Students who want to go but haven't applied:

- Simplify the application process
- Peer mentoring from Group 1 (already gone)
- Early information sessions (Year 1)
- Address financial uncertainty with clear grant info

### Group 4 (resigned non-movers)

Students who don't want to go due to constraint:

- Build international openness through local activities
- Short mobility formats (1–2 weeks)
- Targeted financial support for vulnerability score  $> 6$

### Do not conflate

Deliberate non-movers (free choice) vs resigned non-movers (constrained). Different approaches needed.



Thank you for your attention

Pascaline VINCENT

*Questions, comments, further analyses?*

Full specifications in appendix

Appendix



## Models Estimated

1. **MCA** on barrier items (dimension reduction, 15 items)
2. **Composite score** (MCA Dim1 or simple mean)
3. **OLS** of score on socioeconomic + psychological variables
4. **Logit 1** — Yes/No mobility intention (all 4 groups)
5. **Logit 2** — Applied vs not applied (Groups 2 vs 3)

## Logit specification

$$\text{logit Pr}(Y_i = 1) = \alpha + \sum_j \beta_j Z_{ij}$$

Reported as odds ratios (OR). OR — 1 = % change in odds. McFadden  $R^2 > 0.20$  = good fit.

## Caution

$n = 1230$ : limited power. Group 2 may have  $n < 30$ . Listwise deletion for all models.

## Standardisation formula

All continuous predictors in regressions are standardized:

$$Z_{ij} = \frac{X_{ij} - \bar{X}_j}{\sigma_j}$$

$\bar{X}_j$  and  $\sigma_j$  computed on the analytic sample.

## Variables standardized (logistic models):

- Financial comfort `aisance_fin` (1–5)
- Academic grade `moyenne_acad_norm` (0–10, normalised)
- Parental education `educ_num` (1–6)
- Household income `revenu_num`

## Why standardize?

- Avoids overflow in `glm()` when scales differ
- Makes coefficients comparable: 1 unit = 1 SD shift
- Does **not** affect significance or model fit

## Binary / factor predictors

`depense_imp` (0/1) and `parental_erasmus` (factor) are **not** standardized. Coefficients = log-OR for the reference change (0→1 or No→Yes).

## OLS barrier score model

Predictors **not** standardized in OLS. Coefficients = effect of 1-unit change on original scale.

## Scores & composites

### Intl. profile (0–10):

$$\text{score\_intl} = \frac{\bar{x}_{8 \text{ items}} - 1}{4} \times 10$$
  
Items: travailler\_etr, intl, cultures, tolerance, europe\_id, curiosite, adaptation, monde

**Intl. openness (1–5):** row mean of travailler\_etr, intl, cultures, tolerance, europe\_id

**Personal efficacy (1–5):** row mean of resolution, equipe, adaptation, solutions

### Grade normalisation:

$$\text{acad\_norm} = \frac{\text{grade}}{\text{max}} \times 10$$
  
Florence /30 · Nantes /20 · others /10

## Structural vulnerability (0–10)

$$V_i = (0.4 F_i + 0.3 A_i + 0.3 E_i) / 5 \times 10$$

- $F_i$ : aisance\_fin + revenu\_num + depense\_imp
- $A_i = (10 - \text{acad\_norm}) / 10 \times 5$
- $E_i = (7 - \text{educ\_num}) / 6 \times 5$

Missing → 2.5. Score 0=low, 10=high constraint.

Threshold  $V_i > 6$  = resigned non-mover.

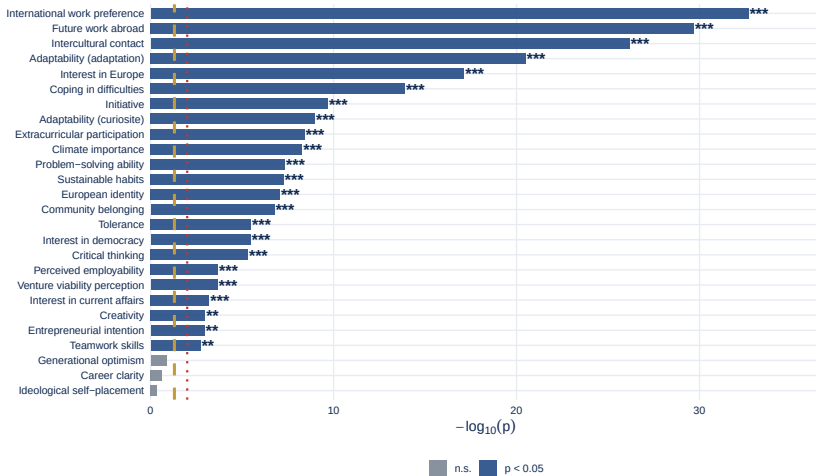
## Z-score reminder

Logistic predictors standardized:

$$Z = (X - \bar{X}) / \sigma.$$

Binary vars (depense\_imp, parental\_erasmus) not standardized.  
OLS uses raw scales.

## Kruskal-Wallis: each psy variable vs 4 groups

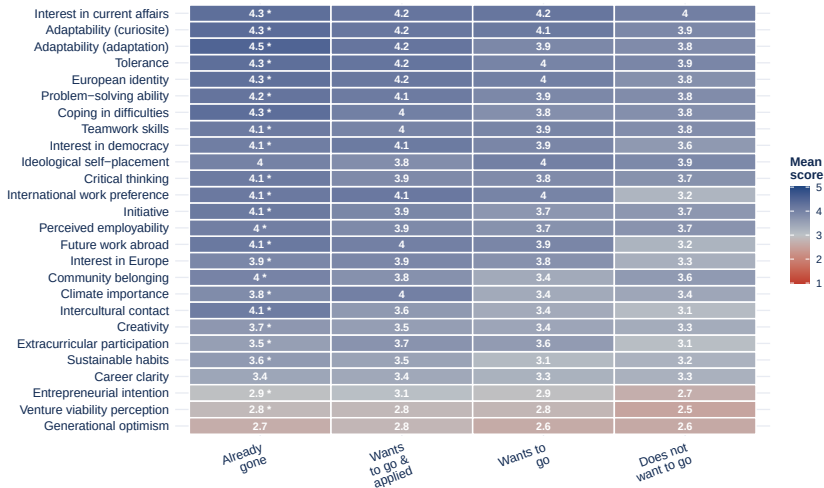
Dashed =  $p=0.05$  --- dotted =  $p=0.01$  -- psy\_science excluded (binary)

23 out of 26 psy variables show significant group differences ( $p < 0.05$ , Kruskal-Wallis). Most discriminating: International work preference, Future work abroad, Intercultural contact. Stars: \*\*\*  $p < 0.001$ , \*\*  $p < 0.01$ , \*  $p < 0.05$ , .  $p < 0.10$ .



### Mean score per psy variable × group (heatmap)

\* in first column = significant Kruskal-Wallis ( $p < 0.05$ ) - psy\_science excluded



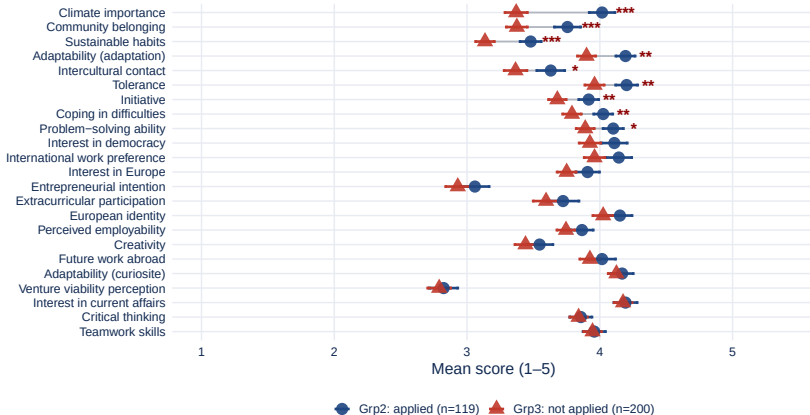
Heatmap: blue = high score (1–5), red = low score. Each cell shows the group mean. Variables marked \* differ significantly across the 4 groups

(Kruskal-Wallis  $p < 0.05$ ). Reading rows: variables with large colour contrast across groups are the most discriminating

# Psy Variables: Groups 2 vs 3 (Significant Only)

## Psy variables: Group 2 vs Group 3

Only variables significant in KW (4 groups) shown · n=319 · \* p<0.10 \*\* p<0.05 \*\*\* p<0.01 (Mann-Wi)



Variables shown: 23 (significant in Kruskal-Wallis across 4 groups). Among these, 7 also differ significantly between Groups 2 and 3 (Mann-Whitney  $p<0.05$ ). Blue circle = Grp2 (applied), red triangle = Grp3 (not applied). Sorted by absolute mean difference – most discriminating at top.